# Failure Analysis Report

FAIL-0229 — Crash

## 1. Failure Summary

| | |
|---|---|
| **Failure ID** | FAIL-0229 |
| **Failure Type** | Crash |
| **Severity** | Trivial |
| **Component** | Push Service |
| **Application** | APP-009 — DeviceHub |
| **Build Version** | 6.2.1 |
| **Timestamp** | 2023-04-30 17:11:23 |
| **Environment** | ENV-002 — Wearable Lab |
| **OS Version** | Wear OS 4 |
| **Network** | Bluetooth 5.3 |
| **Resolution** | Wont Fix |

## 2. Test Case Reference

| | |
|---|---|
| **Test Case ID** | TC-1484 |
| **Title** | Ensure device pairing synchronizes correctly across devices when the server retu... |
| **Execution ID** | EXEC-01070 |
| **Test Run** | RUN-007 — Regression Cycle 7 — Build 5.3.1 |
| **Requirement** | REQ-0391 — Performance — Media Streaming |

## 3. Expected vs Observed Behaviour

| | |
|---|---|
| **Expected Behaviour** | The system should complete media streaming within the defined SLA. |
| **Observed Behaviour** | The system exhibited crash when the user revokes permissions during the operation. |

## 4. Error Details

Error Message:

```
Process com.samsung.smarthome terminated unexpectedly with signal SIGSEGV (segmentation fault).
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.MediaCodec.execute(line 227)
```

| | |
|---|---|
| **Log Reference** | LOG-62076 |

## 5. Executed Steps at Time of Failure

SYNTHETIC DATA — No real Samsung confidential information

| | | |
|---|---|---|
| 1 | Assert that database record for entity 'ENT-754' has status='PEND... | The performance counter logs a value below the SLA threshold. |
| 2 | Enter valid credentials: username='testuser_021', password='[REDA... | The operation completes without error. |
| 3 | Disconnect the wearable and wait 17 seconds. | The audit log records the event with correct metadata. |
| 4 | Simulate Bluetooth interference using the network emulator profil... | A confirmation notification appears on the paired device. |
| 5 | Verify that the UI displays 'Error: retry'. | The operation completes without error. |
| 6 | Disconnect the headset and wait 12 seconds. | The operation completes without error. |

## 6. Root Cause Analysis

Race condition in the connection manager when two threads attempt simultaneous reconnection.

## 8. Historical Similar Failures

| | | | | | | |
|---|---|---|---|---|---|---|
| HIST-0018 | FAIL-0019 | Crash | 5.3.9 | 2021-07-13 | 11 | Wont Fix |
| HIST-0228 | FAIL-0229 | Crash | 6.4.3 | 2022-08-21 | 1 | Resolved |
| HIST-0299 | FAIL-0300 | Crash | 4.1.4 | 2022-02-07 | 3 | Resolved |

## 9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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